MEDICAL DATA ANALYSIS PROJECT

Patient Health Indicators and Simulated Diabetes Risk Patterns

Project Area	Healthcare Data Analytics
Tools Used	Python, Pandas, NumPy, Matplotlib
Dataset	500 synthetic patient-like records
Report Type	Exploratory Data Analysis (EDA)

Prepared as an educational data analysis project

Dataset is synthetic. No real patient information is used.

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1. Executive Summary

This project explores a synthetic healthcare-themed dataset containing 500 records. It examines age, BMI, glucose, blood pressure, cholesterol, physical activity and family history against a simulated two-class outcome label. The analysis uses Python libraries to inspect data quality, replace selected missing numeric values with their median, calculate descriptive statistics and create six visualizations.

After cleaning, the dataset contains 500 records. There are 29 records with the simulated higher-risk label (5.8%) and 471 records with the lower-risk label (94.2%). These are generated labels for a classroom exercise, not confirmed diagnoses.

Important safety note: All records and labels are artificially generated. The project is for learning data analysis only and must not be used for diagnosis, screening, treatment or real-world clinical decisions.

2. Project Prompt and Objectives

Project prompt

You are a junior data analyst supporting a fictional health research team. The team wants a simple exploratory report describing a synthetic patient dataset and comparing health indicator patterns between two simulated outcome groups. Your task is to load, clean, analyze and visualize the data, then explain the findings in accessible language.

Objectives

• Understand the structure and variables in a healthcare-themed dataset.

• Identify missing values and duplicate records.

• Prepare a clean dataset using a transparent, repeatable method.

• Summarize the distribution of key numeric health indicators.

• Compare averages and distributions across simulated outcome groups.

• Communicate results clearly while avoiding unsupported medical claims.

3. Dataset Overview and Data Dictionary

The project uses a generated dataset of 500 fictional records. Each row represents one artificial record. The Excel appendix file includes the full data, a data dictionary and a read-me sheet. The CSV file is provided for direct use in the notebook.

Variable	Meaning	Type
Patient_ID	Artificial record identifier	Text
Age	Age in years	Integer
BMI	Body Mass Index (kg/m²)	Decimal
Glucose_mg_dL	Glucose measurement (mg/dL)	Integer
Blood_Pressure_mmHg	Blood pressure value (mmHg)	Integer
Cholesterol_mg_dL	Cholesterol measurement (mg/dL)	Integer
Physical_Activity_hours_week	Activity hours per week	Decimal
Family_History	1 = yes; 0 = no	Binary
Diabetes_Status	Simulated outcome label: 1 or 0	Binary

The outcome column is a simulated teaching label created by a mathematical rule with random variation. It is not a clinically validated outcome.

4. Tools and Methodology

Tools

• Python: main programming language.

• Pandas: reading, cleaning and summarizing tabular data.

• NumPy: numerical operations and dataset generation.

• Matplotlib: charts and visual exploration.

• Jupyter Notebook: step-by-step, reproducible analysis.

Analysis workflow

Step	Action
1	Load the CSV file and inspect rows, columns and data types.
2	Check missing values and duplicate records.
3	Fill missing numeric values with each column's median.
4	Calculate descriptive statistics and group averages.
5	Create six charts to explore distributions and relationships.
6	Summarize patterns and state limitations.

5. Data Cleaning and Preparation

Column	Missing values
BMI	7
Cholesterol_mg_dL	6
Physical_Activity_hours_week	5
Duplicate rows	0

Missing values were intentionally introduced in BMI, cholesterol and weekly physical activity to demonstrate a common cleaning task. The median of each affected column was used to fill missing values. Median imputation is simple and less

affected by extreme values than the mean. No duplicate records were found in the generated dataset.

The family-history and outcome fields were kept as integer binary values (0/1). The original dataset remains unchanged; cleaning is performed on a copy so that the raw data is preserved.

6. Exploratory Analysis and Results

6.1 Simulated outcome label distribution

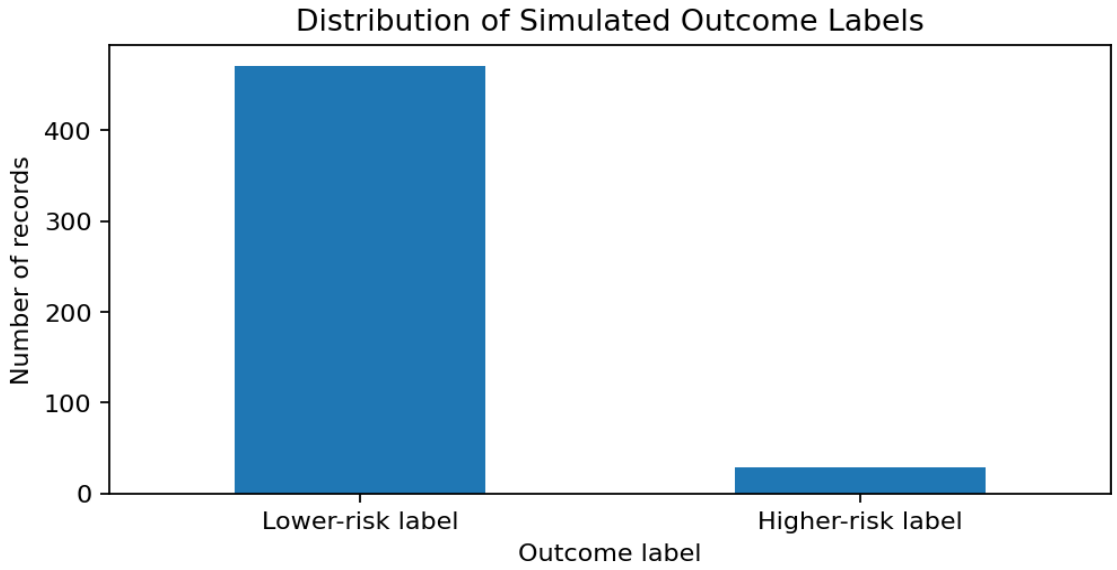


Figure 1. Count of each generated label. Higher-risk label: 29; lower-risk label: 471.

The chart shows how the 500 generated records are divided between the two labels. The higher-risk label represents 5.8% of this artificial dataset. This percentage is not a population prevalence estimate.

6.2 Age distribution

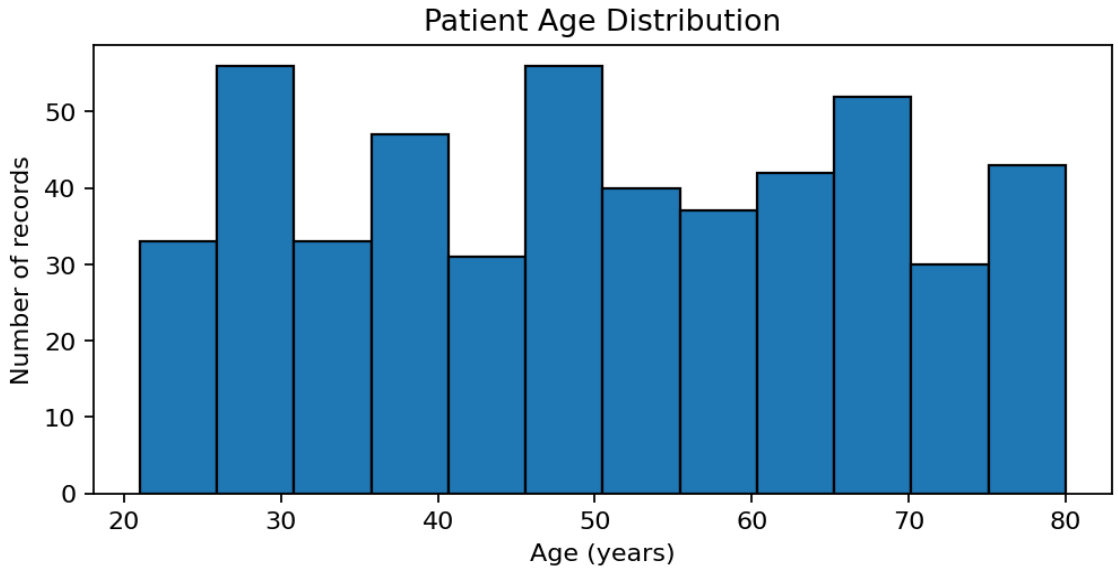


Figure 2. Histogram showing the number of records across age ranges.

Age ranges from 21 to 80 years in this dataset. The histogram helps identify which age bands are more represented. It does not show disease incidence or age-specific medical risk.

6.3 Glucose distribution

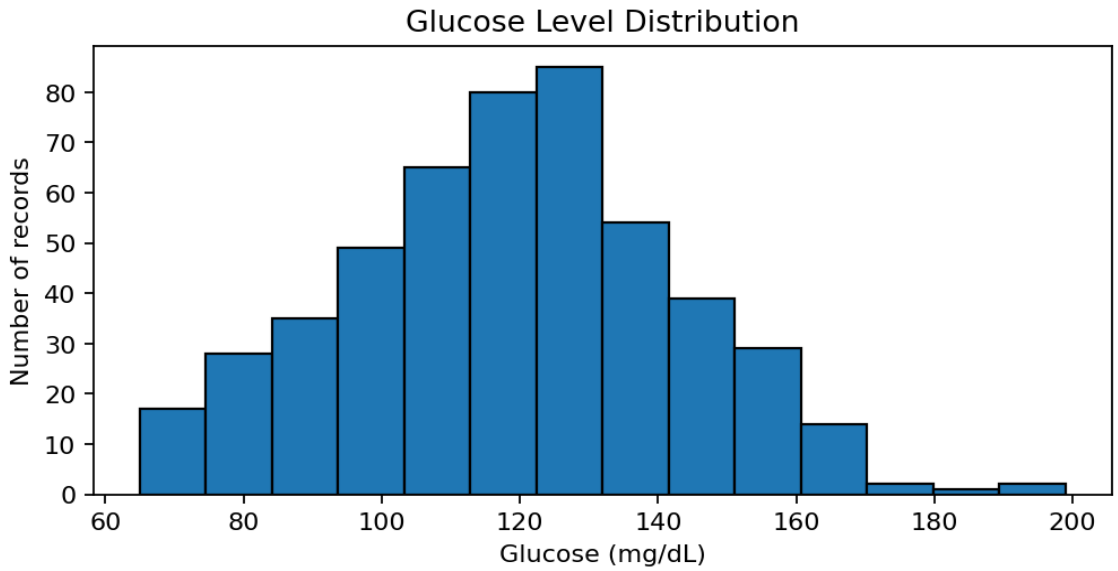


Figure 3. Histogram of glucose measurements in the synthetic records.

Glucose values range from 65 to 199 mg/dL. The histogram shows the spread of generated values and whether observations cluster around particular ranges. These values are simulated and should not be interpreted using clinical thresholds.

6.4 BMI comparison by label

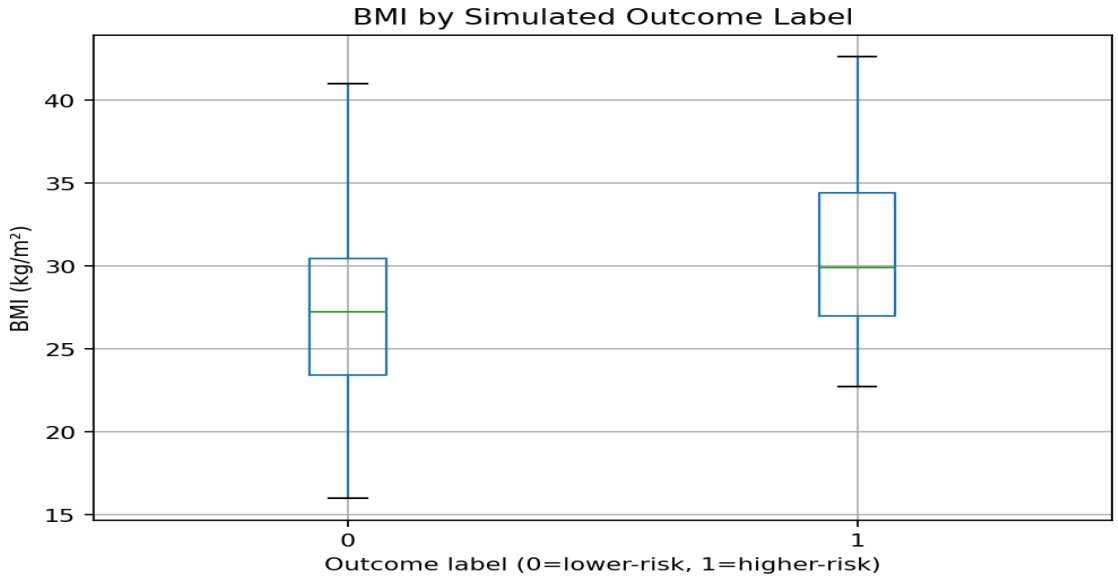


Figure 4. Box plot comparing BMI distributions for the two simulated outcome labels.

The box plot compares the median, spread and possible outliers in BMI for each label. Any difference is descriptive only. It does not establish that BMI caused the generated label or predicts an individual's health.

6.5 Average glucose by label

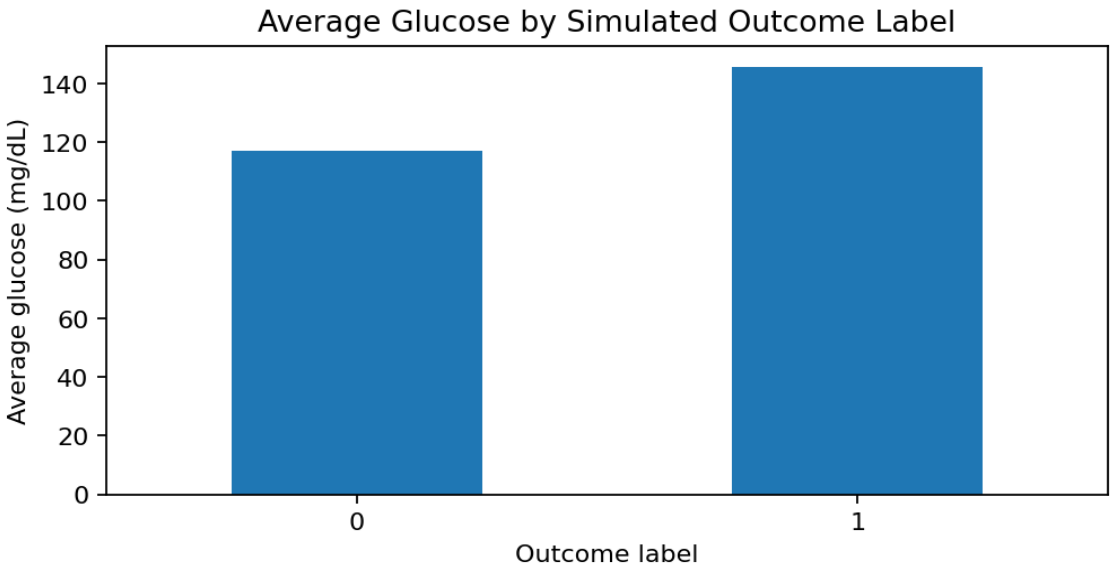


Figure 5. Mean glucose measurement for each simulated outcome group.

This chart compares average glucose values between the two groups. The label-generation process uses glucose as one of several inputs, so a group difference is expected in this synthetic exercise. It should not be treated as evidence from a real clinical study.

6.6 Correlation heatmap

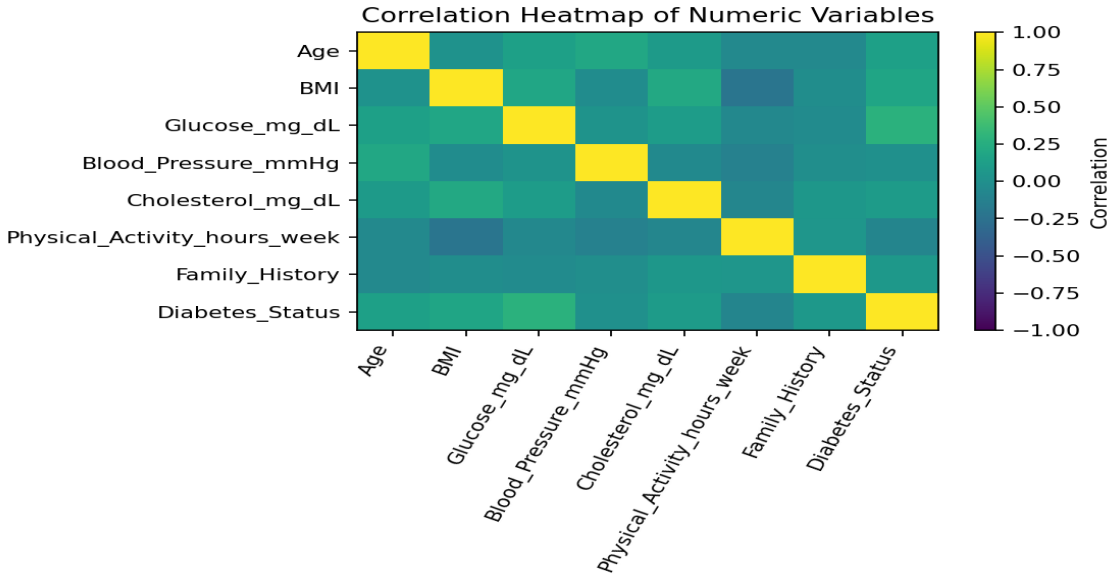


Figure 6. Correlation matrix for numeric variables. Values closer to +1 or -1 indicate stronger linear association.

The heatmap provides a quick view of linear associations among numeric columns. Correlation does not prove causation, and relationships in generated data may reflect the formula used to create the dataset.

7. Key Findings and Recommendations

• The dataset contains 500 artificial records and 9 columns.

• The simulated higher-risk label appears in 29 records (5.8%).

• The median age is 50.0 years.

• The average BMI is 27.29 kg/m².

• The average glucose measurement is 118.65 mg/dL.

• The median weekly physical activity is 4.3 hours.

Recommendations for a student analysis project:

• Keep the raw dataset and cleaned dataset separate for auditability.

• Document every cleaning decision and check whether imputation changes distributions.

• Use labels and units on every chart so that results are easy to understand.

• For future work, practice train/test splitting and model evaluation only with a properly sourced, ethically approved and clinically reviewed dataset.

• Avoid making diagnosis, treatment or individual risk claims from exploratory charts.

8. Limitations and Ethical Considerations

• Synthetic data: all records are artificially generated; they do not represent real people.

• Simulated label: Diabetes_Status is not a medical diagnosis and is not clinically validated.

• Simplified variables: the dataset excludes many relevant clinical, lifestyle and demographic factors.

• No causal inference: charts and correlations describe patterns, not causes.

• No clinical use: the project must not be used for screening, diagnosis, treatment or health decisions.

Ethical note: real medical data requires privacy safeguards, informed permissions where applicable, secure storage, and review by appropriate institutional and clinical authorities.

9. Conclusion

The project demonstrates the main stages of a beginner-friendly data analysis workflow using a healthcare-themed dataset. It includes data inspection, missing-value handling, descriptive statistics, group comparisons and six visualizations. The exercise shows how charts can communicate patterns, while also demonstrating why analysts must explain data quality, assumptions and limitations. Since the dataset and labels are synthetic, the findings are suitable only for classroom learning.

Appendix A: Dataset Sample and File Details

The complete dataset is supplied separately as `Medical_Dataset.xlsx` and `Medical_Dataset.csv`. The Excel workbook contains three sheets: `Medical_Data` (all 500 records), `Data_Dictionary` (column explanations) and `Read_Me` (important usage notes).

Patient_ID	Age	BMI	Glucose_mg_dL	Blood_Pressure_mmHg	Cholesterol_mg_dL	Physical_Activity_hours_weekly	Family_History	Diabetes_Status
P0001	26	27.0	141	64	229.0	2.7	0	1
P0002	67	18.4	89	99	202.0	2.4	0	0
P0003	60	19.9	69	73	143.0	6.5	0	0
P0004	47	38.6	131	89	224.0	4.2	0	0
P0005	46	20.8	133	78	156.0	3.6	1	0
P0006	72	21.8	161	76	173.0	1.8	0	0
P0007	26	37.1	134	55	204.0	2.1	0	0
P0008	62	42.6	128	78	197.0	2.9	0	1

Table A1. First eight rows of the supplied raw dataset. The Excel and CSV files contain the complete data.

Appendix B: Reproducibility

To reproduce the analysis:

• Keep `Medical_Data_Analysis.ipynb` and `Medical_Dataset.csv` in the same folder.

• Open the notebook in Jupyter Notebook, JupyterLab or Google Colab (upload both files).

• Run the cells from top to bottom.

• Required Python packages: pandas, numpy and matplotlib.

Project files supplied: `Medical_Data_Analysis.ipynb`, `Medical_Dataset.csv`, `Medical_Dataset.xlsx`, and this PDF report.

End of report